

Figure placement proof (not the manuscript)

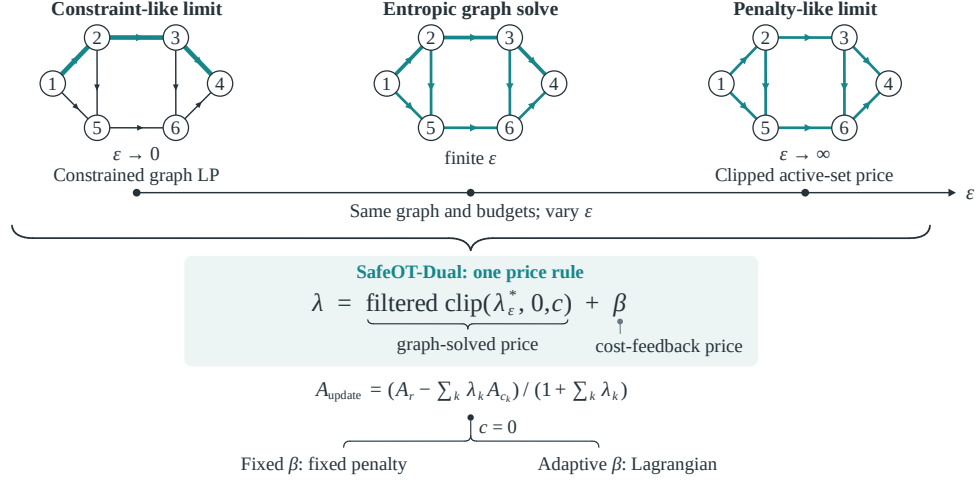


Figure 1: One price rule across safe-RL regimes. SafeOT-Dual combines a clipped, filtered graph-solved price with realized-cost feedback. Under the endpoint assumptions, the graph problem approaches a constrained LP as $\epsilon \rightarrow 0$; at large ϵ , a positive limiting information-projection multiplier yields a clipped active-set price. Clipping bounds the graph contribution in both regimes. At cap $c = 0$, fixed β gives a fixed penalty and adaptive β gives a Lagrangian rule. These regimes do not imply exact recovery of external constrained-policy algorithms. Miniatures are partial schematic graphs on shared support, not measured or complete balanced-flow solutions.

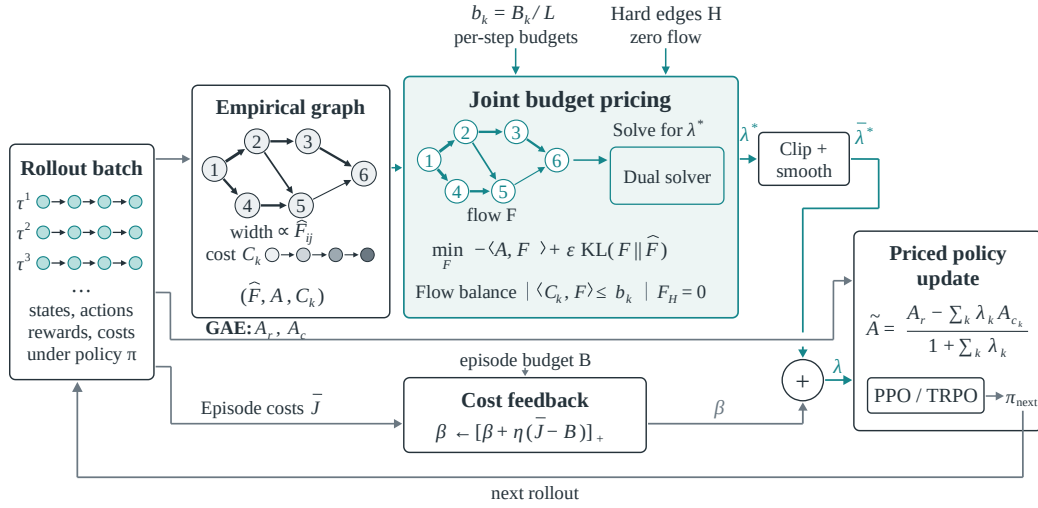


Figure 2: On-policy SafeOT-Dual. Each rollout supplies an empirical graph, reward/cost GAE advantages and episode-cost feedback. Joint graph pricing uses discounted flow balance, per-step soft budgets $b_k = B_k/L$ and hard-edge support to obtain λ^* . Only prices transfer: after clipping and filtering, the graph price combines with feedback integrator β in a normalized priced-advantage PPO/TRPO step. GAE follows a separate branch; the new policy generates the next rollout. Graph feasibility alone does not certify policy safety. The separate off-policy executor is not shown.